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Brando Miranda

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Google Scholar; Website; Stanford Profile; MIT Website

RESEARCH INTERESTS

AI for formal mathematics and Lean 4 theorem proving — autoformalization, end-to-end formal verification of AI-generated code, and contamination-resistant evaluation of LLM mathematical reasoning. Lead/co-lead on Lean 4 benchmarks and tools: *Putnam-AXIOM* (ICML 2025), *VeriBench* (2nd AI for Math Workshop @ ICML 2025), *Pantograph* (TACAS 2025), and an autoformalization benchmark (ICLR 2024 Tiny Papers Track). Also contributed to *Morph Prover v0-7B* (first frontier Lean 4 model) and *Moogles.ai* (Lean-code retrieval/search system). Adjacent data-centric ML work — NeurIPS Outstanding Paper on the “emergent-abilities mirage” and the *Beyond Scale* diversity coefficient — directly informs corpus design and contamination-resistant evaluation for AI mathematics.

EDUCATION

Ph.D. in Computer Science

Stanford University

2022-2026 (*expected*)

GPA: 4.045/4.0

Advisor: Prof. Sanmi Koyejo, Stanford Trustworthy AI Research (STAIR) group

Fellowships: Stanford School of Engineering Fellowship, EDGE Scholar

Master of Engineering in Electrical Engineering and Computer Science

Massachusetts Institute of Technology

2014-2016

GPA: 4.8/5.0

Advisor: Prof. Tomaso Poggio, Center for Brains, Minds and Machines (CBMM)

Focus: Deep learning theory

Bachelor of Science, Computer Science and Engineering

Massachusetts Institute of Technology

2010-2014

minor in Mathematics & Music

AWARDS & HONORS

- **NeurIPS Outstanding Main Track Paper Award** (December 2023) (top 0.4%, only 2 papers selected)
- **ICML Outstanding Paper TiFA Workshop Award** (July 2024)
- **EDGE Scholar**, Stanford University (September 2022)
- **Stanford School of Engineering Fellowship** (September 2022)
- **Honorable Mention**, Ford Foundation Fellowship (2020, 2021)
- **Best Research Project Award**, UIUC graduate course CS 598 “Learning to Learn” (December 2020)
- **Most Cited Paper Certificate**, International Journal of Automation & Computing (IJAC) (December 2019)
- **Computer Science Excellence Saburo Muroga Endowed Fellow**, UIUC (2019-2020)
- **Sloan Scholar**, Alfred P. Sloan Foundation’s Minority Ph.D. (MPHD) Program (2018-2019)
- **Grainger Engineering SURGE Fellowship** (2018-2019)

SELECTED PUBLICATIONS

VeriBench: End-to-End Formal Verification Benchmark for AI Code Generation in Lean 4 (2025)

Brando Miranda, Zhangir Zhou, Anima Nie, Elío Obbad, Leni Aniva, Kai Fronsdal, William Kirk, Dogus Soyulu, et al.

2nd AI for Math Workshop @ ICML 2025

CoDaPO: Confidence and Difficulty-Adaptive Policy Optimization for Post-Training Language Models (2025)

Zhangir Zhou, Xingrun Lu, Cheng Cao, Brando Miranda, Tong Liu, Bo Han, Sanmi Koyejo

2nd AI for Math Workshop @ ICML 2025

Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive? (2025)

Rylan Schaeffer, Hailey Schoelkopf, Brando Miranda, Gabriel Mukobi, Varun Madan, Adam Ibrahim, Herbie Bradley, Stella Biderman, Sanmi Koyejo

International Conference on Machine Learning (ICML) 2025 & ICML TiFA Workshop Outstanding Paper Award 2024

[arXiv]

Putnam-AXIOM: A Functional & Static Benchmark for Measuring Higher Level Mathematical Reasoning in LLMs (2025)

Aryan Gulati, Brando Miranda, Eric Chen, Emily Xia, Kai Fronsdal, Bruno de Moraes Dumont, Sanmi Koyejo

International Conference on Machine Learning (ICML) 2025 & NeurIPS MATH-AI Workshop 2024

Pantograph: A machine-to-machine interaction interface for advanced theorem proving, high level reasoning, and data extraction in Lean 4 (2025)

Leni Aniva, Chuyue Sun, Brando Miranda, Clark Barrett, Sanmi Koyejo

International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS) 2025

[Springer Link]

Causally Quantifying the Effect of Test Set Contamination on Generative Benchmarks (2025)

Rylan Schaeffer, Brando Miranda, Jiaqi Kazdan, Kaivu Liu, Amir M. Ahmed, Niloofar Mireshghallah, et al.

NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle

Position: Machine Learning Conferences Should Establish a ‘Refutations and Critiques’ Track (2025)

Rylan Schaeffer, Jiaqi Kazdan, Yarin Denisov-Blanch, Brando Miranda, Max Gerstgrasser, et al.

Preprint 2025

Failures to Find Transferable Image Jailbreaks Between Vision-Language Models (2024)

Rylan Schaeffer, Dan Valentine, Luke Bailey, James Chua, et al., Brando Miranda, et al., Sanmi Koyejo, Ethan Perez

International Conference on Learning Representations (ICLR) 2024 & NeurIPS Red Teaming GenAI Workshop 2024

[OpenReview]

Does Maximizing Neural Regression Scores Teach Us About The Brain? (2024)

Rylan Schaeffer, Mikail Khona, Sankarshan Chandra, Michael Ostrow, Brando Miranda, Sanmi Koyejo

UniReps Workshop 2nd Edition, NeurIPS 2024

Quantifying the Importance of Data Alignment in Downstream Model Performance (2024)

Keshav Chawla, Aditya Sahai, Marco DePavia, Sathvik Sundar, Brando Miranda

ICLR 2024 Workshop on Data-centric Machine Learning Research (DMLR)

An Evaluation Benchmark for Autoformalization in Lean4 (2024)

Aryan Gulati, Devanshu Ladsaria, Shubham Mishra, Jasdeep Sidhu, Brando Miranda

The Second Tiny Papers Track at ICLR 2024

Are Emergent Abilities of Large Language Models a Mirage? (2023)

Rylan Schaeffer, Brando Miranda, Sanmi Koyejo

Neural Information Processing Systems (NeurIPS) Outstanding Main Track Paper Award & Oral 2023

Featured in: New York Times, Quanta Magazine, Forbes, Stanford HAI

Cited in the 2024 Economic Report of the President (White House)

[arXiv] [OpenReview]

Morph Prover v0 7b: The 1st Frontier Model for the Lean 4 Formal Verification Programming Language (2023)

Brando Miranda (Machine Learning Research Scientist Consultant) & Morph Labs Team

[Blog] [Hugging Face Model Card]

Beyond Scale: the Diversity Coefficient as a Data Quality Metric Demonstrates LLMs are Pre-trained on Formally Diverse Data (2023)

Alycia Lee, Brando Miranda*, Patrick Yu, Oluwasanmi Koyejo (*equal contribution)*

ICML Data-Centric Machine Learning Workshop & ICML Deployable Generative AI Workshop 2023

[arXiv] [Code]

Why and when can deep-but not shallow-networks avoid the curse of dimensionality: a review (2017)

Tomaso Poggio, Hrushikesh Mhaskar, Lorenzo Rosasco, Brando Miranda, Qianli Liao

International Journal of Automation and Computing 2017, Most Cited Paper Certificate

[Journal Link]

Google Scholar metrics: 2,948+ citations — h-index: 17 — i10-index: 22

RESEARCH & PROFESSIONAL EXPERIENCE

Stanford University - Stanford, CA

September 2022 - 2026 (expected)

Ph.D. Student in Computer Science

- Ph.D. research with Prof. Sanmi Koyejo on **AI for formal mathematics and Lean 4**: benchmarks, tooling, and training methods for autoformalization, theorem proving, and end-to-end formal verification of AI-generated code
- Lead/co-lead on four Lean 4 artifacts: *Putnam-AXIOM* (functional/static Putnam-level math reasoning, ICML 2025), *VeriBench* (end-to-end Lean 4 formal verification, 2nd AI for Math Workshop @ ICML 2025), *Pantograph* (machine-to-machine Lean 4 interface, TACAS 2025), and an autoformalization benchmark for Lean 4 (ICLR 2024 Tiny Papers Track)
- Founding member of Stanford AI for Lean, a Stanford research community advancing AI for Lean theorem proving and autoformalization of mathematics
- Published award-winning work challenging prevailing notions of “emergent abilities” in large language models (NeurIPS Outstanding Paper Award 2023); data-centric ML work (*Beyond Scale* diversity coefficient; contamination quantification) directly informs Lean autoformalization corpus design and benchmark integrity
- Research mentor for undergraduate and master’s students focused on data quality for Foundation Models

Morph Labs - Remote

2023

Machine Learning Research Scientist Consultant

- **Lean 4 frontier model**: co-built *Morph Prover v0-7B*, the first openly released frontier model trained for the Lean 4 formal verification language
- **Lean retrieval**: led the embedding system behind *MoogLe.ai* for verified Lean-code search, enabling premise selection and proof retrieval workflows

Wise Agents - Stanford Spin-out

2023

AI Research Consultant

- Consulted on AI agent-based systems to transform sales performance
- Company featured in Forbes Mexico

IBM Research - Yorktown Heights, NY

May 2022 - August 2022

Graduate Research Intern

- Developed transformer models to predict mathematical formal proofs using the Coq theorem proving language

University of Illinois Urbana-Champaign - Urbana-Champaign, IL

September 2018 - 2021

Ph.D. Student in Computer Science

- Conducted foundational research on meta-learning, transfer learning, and data diversity
- Published multiple papers on the relationship between task diversity and learning performance

IBM Research - Yorktown Heights, NY

May 2021 - August 2021

Graduate Research Intern

- Studied optimization and architectural inductive biases of Transformer models for program type inference

MIT CBMM (Center for Brain Minds & Machines) - Cambridge, MA

June 2015 - September 2018

Research Assistant

- Conducted deep learning theory research with Prof. Tomaso Poggio (who also mentored DeepMind co-founder Demis Hassabis)
- Contributed to highly-cited work on generalization theory and function approximation for neural networks

MEDIA COVERAGE

- **White House Economic Report of the President (2024)**: Our work on emergent abilities was cited in the official economic report to the President
- **The New York Times (2023)**: “Silicon Valley Confronts the Idea That the ‘Singularity’ Is Here”
- **Quanta Magazine (2024)**: “How Quickly Do Large Language Models Learn Unexpected Skills?”
- **Forbes (2023)**: “AI ‘Emergent Abilities’ Are A Mirage, Says AI Researcher”
- **Andrew Ng (2024)**: Endorsed our paper as evidence that AGI development will be smooth and predictable
- **Additional coverage**: Stanford HAI, Y Combinator News, Vice, Medium, HackerNews, NeurIPS blog

TEACHING EXPERIENCE

University of Illinois Urbana-Champaign

August 2020 - December 2020

Graduate Teaching Assistant

- CS 446 Machine Learning
- Designed problem sets and exams; held weekly office hours

Massachusetts Institute of Technology

2014-2016

Graduate Teaching Assistant

- Statistical Learning Theory & Applications (9.520/6.860) with Prof. Lorenzo Rosasco & Tomaso Poggio
- Introduction to Algorithms (6.006) with Prof. Nancy Lynch, Bruce Tidor and Aleksander Madry
- Design & Analysis of Algorithms (6.046) with Prof. Shafi Goldwasser, Dana Moshkovitz, Nir Shavit
- Introduction to Machine Learning (6.036) with Prof. Tommi Jaakkola, Suvrit Sra & Regina Barzilay

LEADERSHIP & SERVICE

Research Mentorship

2016 - Present

- Mentored undergraduate and graduate students at Stanford, UIUC, and MIT CBMM
- Led research teams on data quality metrics, meta-learning, and formal reasoning

Academic Service

2018 - Present

- Reviewer for ICLR 2020, JMLR 2018
- Graduate advisor for Latinos in Computer Science (LCS) at UIUC
- Founding member of Stanford AI for Lean, a research community advancing AI for Lean theorem proving and formalizing mathematics
- Founded "Stanford Bachata Sensual & Brazilian Zouk" and "UIUC Bachata Sensual & Zouk" official student organizations

Outreach

2017 - Present

- Distributed Research Experiences for Undergraduates (DREU) UIUC 2019
- Undergraduate Research Opportunity Program (UROP) MIT 2017-2018
- Engineering of Intelligence Team (EIT) CBMM MIT 2017-2018

SKILLS & TOOLS

Programming: Python, PyTorch, TensorFlow, JAX, Lean 4, OCaml, Coq

ML/AI Tools: Hugging Face, Weights & Biases, DSPy, git, UNIX

Languages: English, Spanish (native fluency)